

Revenue Leakage Analysis | October 2024

Exploratory Data Analysis · SQL · Power BI

PROBLEM STATEMENT

The Souled Store processed **60,000 orders** totalling **₹62.46M GMV** in October 2024, yet a meaningful slice of that revenue never converted to profit. This analysis quantifies where value is being lost — through cancellations, COD friction, aggressive discounting, and high-return SKUs — and surfaces actionable levers to recover margin.

DATASET OVERVIEW

Metric	Value	Metric	Value
Total Orders	60,000	Products	80 SKUs
Gross GMV	₹62.46M	Franchises / Themes	14
Avg Order Value	₹1,041	Period	Oct 1–31, 2024
Cities Covered	Pan-India	Payment Methods	6 (COD, UPI, etc.)

KEY FINDINGS

01 ₹5.35M in Cancelled Order Value Total revenue lost to order cancellations reached ₹5.35M across 5,000 cancelled orders — an 8.54% overall cancellation rate. COD alone accounts for 22.9% of that loss (₹1.23M), despite representing only 15.7% of total order volume.	02 Discount Erosion: 88.6% of Orders Carry a Discount 51,647 of 60,000 orders had a discount applied, with an average discount of 15.3% (₹144.73/order). Total discount outlay hit ₹8.68M. Gross margin held at 31.47%, but 30%+ discount bands show the sharpest margin compression.
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03

Instagram Drives 15.9M — 4.2× Direct Traffic

Instagram is the #1 revenue channel at 15.9M, followed by Influencer Marketing (10.9M) and Facebook Ads (9.6M). Organic Search contributes 7.1M. Direct Traffic lags at 3.8M, signalling low brand-recall shopping behaviour.

04

Heritage Sneaker Low Top: 4.85M — but High Return Risk

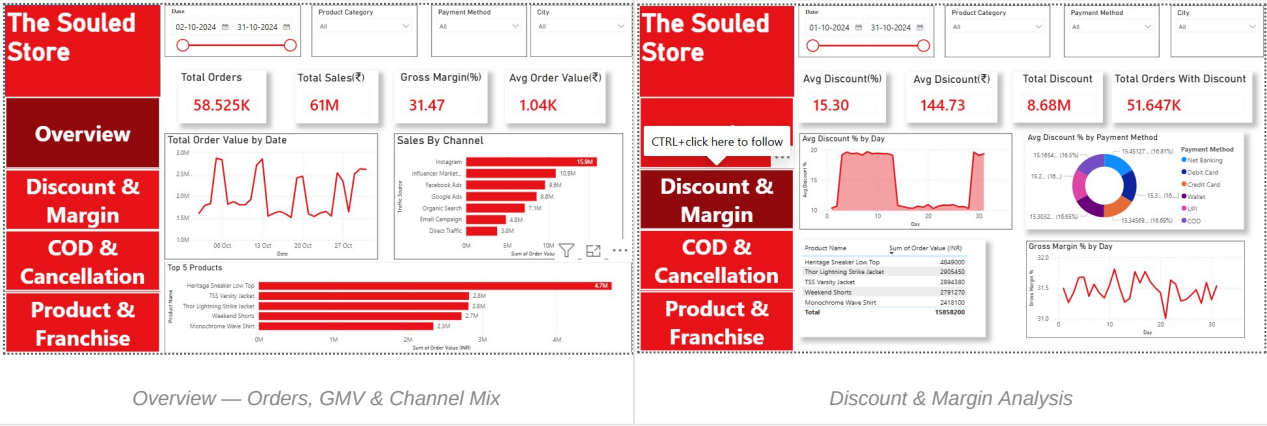
The top revenue SKU (Heritage Sneaker Low Top, 4.85M) and Thor Lightning Strike Jacket (2.9M) also rank among the highest-volume products. Return rate analysis flags that revenue leaders don't always have the cleanest post-purchase metrics.

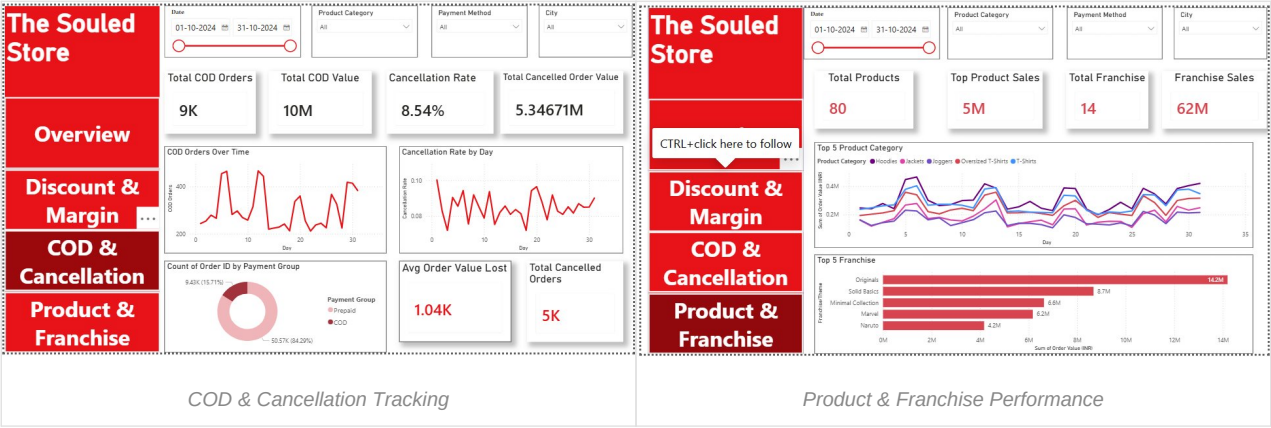
05

Originals Franchise Dominates at 14.2M

The Originals franchise leads all themes at 14.2M, followed by Solid Basics (8.7M) and Minimal Collection (6.6M). Marvel and Naruto IP franchises each clear 4–6M, confirming licensed content as a consistent revenue driver alongside house brands.

POWER BI DASHBOARD PREVIEW





SQL HIGHLIGHTS

COD Cancellation Rate by City

Identifies which cities have the worst COD fallout — critical for logistics risk tiering.

```
SELECT City,
       COUNT(*) AS Total_COD_Orders,
       ROUND(100.0 * SUM(CASE WHEN Order_Status = 'Cancelled'
                              THEN 1 ELSE 0 END) / COUNT(*), 2) AS Cancellation_Rate
FROM SOULED_STORE
WHERE Payment_Method = 'COD'
GROUP BY City
ORDER BY Cancellation_Rate DESC;
```

Discount Band vs. Margin

Quantifies margin erosion at each discount tier — the core evidence for discount cap policy.

```
SELECT
  CASE WHEN Discount_Applied = 0 THEN '0%'
        WHEN Discount_Applied BETWEEN 1 AND 10 THEN '1-10%'
        WHEN Discount_Applied BETWEEN 11 AND 20 THEN '11-20%'
        ELSE '20%+' END AS Discount_Band,
  COUNT(*) AS Order_Volume,
  ROUND(AVG(Estimated_Profit_Margin), 2) AS Avg_Margin
FROM SOULED_STORE
GROUP BY Discount_Band ORDER BY Avg_Margin DESC;
```

Delivery Partner Rating & Return Correlation

Links delivery partner performance (speed, rating) to post-delivery return rates — reveals operational NPS risk.

```
SELECT Delivery_Partner,
       ROUND(AVG(Customer_Rating), 2) AS Avg_Rating,
       ROUND(AVG(Delivery_Time_Days), 2) AS Avg_Delivery_Days,
       ROUND(100.0 * SUM(CASE WHEN Return_Request = 'Yes'
                              THEN 1 ELSE 0 END) / COUNT(*), 2) AS Return_Rate
FROM SOULED_STORE
WHERE Customer_Rating IS NOT NULL
GROUP BY Delivery_Partner
ORDER BY Avg_Rating ASC;
```

RECOMMENDATIONS

01 Cap COD availability to cities below 6% cancellation rate

COD accounts for █1.23M in cancellation losses despite being only 15.7% of orders. Introduce a city-level COD eligibility score: restrict or add a nominal prepayment deposit (█50–100) for high-COD-cancellation pin codes. This alone could recover █400–600K annually with minimal checkout friction.

02 Enforce a 20% discount ceiling on Originals & high-margin SKUs

Originals franchise drives █14.2M in revenue and is the brand's most elastic property. The current 15.3% average discount with █8.68M total outlay suggests discounting is being applied bluntly rather than strategically. Reserve 21%+ discounts for clearance inventory and end-of-season events only — protecting the 31.47% gross margin baseline.

03 Double down on Instagram + Influencer; audit Google Ads ROI

Instagram (█15.9M) and Influencer Marketing (█10.9M) together drive 43% of GMV. Facebook Ads (█9.6M) and Google Ads (█8.6M) add volume but at higher paid spend. Direct Traffic (█3.8M) indicates low brand loyalty — invest in post-purchase CRM flows (email/WhatsApp) to convert first-time buyers into repeat customers, especially given the █1,041 AOV opportunity.

TOOLS USED

Tool	Purpose	Key Output
Python (pandas, seaborn)	Exploratory Data Analysis & visualisation	EDA notebook with 15+ charts
SQL (Standard)	Aggregation, window functions, cohort queries	5 analytical queries for business KPIs
Power BI	Interactive dashboard across 4 analytical pages	Filterable report with slicers + KPI cards

Dataset is synthetic and self-generated for portfolio purposes. All figures, order records, and customer data are fictional and do not represent actual Souled Store operations or financials. Prepared as part of a Data Analyst application.